

Element C

Justification of Design Specifications

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Problem Statement

Medical institutions lack a cost-effective, risk-free method for surgical training and other medical processes. Certified cadavers are the best training material for students, but are in great shortage worldwide, and pose ethical and practical issues. Due to said scarcity, organizations such as government agencies allocate a great proportion of their budget towards acquiring these materials: Medicare spends ~\$180,000 per medical resident each year through subsidies to universities, totaling over \$25 billion.¹ Overall, inefficient distribution and lack of product availability hinder surgical trainees worldwide, thus a more accessible solution is required.

Introduction

To design an effective solution for the problem, specifications must be in place to ensure a quality product that meets consumer needs. Determining testable criteria involved comprehensively analyzing current market products, and utilizing a similar solution matrix to determine common strengths and weaknesses. This also included researching gaps in the market to specify an effective solution. These design requirements are ranked in order of prevalence, the first being of utmost importance. Comparing such allows for an understanding of which needs should be met first.

I. Primary Stakeholders

The target market for a digital VR device for surgical training would be medical institutions, including university training programs and medical schools. These organizations are the main users of educational cadavers and would benefit the most from this technology to reduce the need for such. However, many hands-on technical fields that require physical resources could also benefit from the product such as construction, welding, and piloting work. Colleges worldwide that specialize in engineering, aviation, and metalwork would use this product to remove reliance on physical necessities.

¹ "The Costs of Training Internal Medicine Residents in the United States." The American Journal of Medicine, U.S. National Library of Medicine, pubmed.ncbi.nlm.nih.gov/25063649/. Accessed 12 Nov. 2024.

II. Specifications

1. Restrictive Force on Fingers

- a. Specification: The solution shall exert a scalable force equal and opposite to the force applied by the fingers, with a maximum of 8 lbf.
- b. Rationale: The average human can apply 35N (~7.9lbf) of force with one finger². To replicate physical solid touch, the motors must be able to overpower the fingers within a set range of forces.

2. Position and Rotational Accuracy

- a. Specification: The solution shall track the positions of each finger to a sub 5mm accuracy, and the angle of each finger to sub 5° accuracy.
- b. Rationale: Similar solutions such as the Meta Quest 2's hand tracking have an accuracy of 1.1cm and a rotation accuracy of ~10 degrees³. Medical practices require precision, thus the product shall have less margin of error than comparable market products.

3. Element Resistance and Longevity

- a. Specifications: The solution shall stay below 36 °C in use, and withstand prolonged ambient temperatures, different humidities, and water/sweat precipitation.
- b. Rationale: The solution cannot get too hot to prevent uncomfortable heat for the user. Additionally, corrosion and water resistance should be added to ensure excessive dust or sweat does not compromise the functionality of the solution, helping to prolong product lifespan.⁴

4. Weight

- a. Specifications: The solution shall weigh at most 400 grams.
- b. Rationale: Practice surgery sessions can take many hours, and a lightweight solution allows for more natural and unobtrusive usage. Market solutions such as the HaptX Gloves G1 weigh 450g,⁵ hence a comparable yet lighter weight is best.

² Riddle, Michael et al. (2020). Evaluation of individual finger forces during activities of daily living in healthy individuals and those with hand arthritis. *Journal of Hand Therapy*, 33(2), 188-197.
[https://www.jhandtherapy.org/article/S0894-1130\(20\)30079-X/abstract](https://www.jhandtherapy.org/article/S0894-1130(20)30079-X/abstract)

³ AbdIkrim, D., Di Luca, M., Aves, P., Maaroufi, M., Yeo, S. H., Miall, R. C., Holland, P., & Galea, J. M. (2024). A methodological framework to assess the accuracy of virtual reality hand-tracking systems: A case study with the Meta Quest 2. *Behavior research methods*, 56(2), 1052–1063. <https://doi.org/10.3758/s13428-022-02051-8>

⁴ Gelston, Jon. *When Should You Waterproof Your Electronics Devices? Here's An Answer.*, www.aimprocessing.com/blog/when-should-you-waterproof-your-electronics-devices-heres-an-answer. Accessed 14 Dec. 2024.

⁵ HaptX. (2024). *HaptX Gloves G1*. HaptX. <https://haptx.com/gloves-g1/>

5. Joint Materials

- a. Specifications: The finger connection points shall be made of a flexible material.
- b. Rationale (Self-Imposed): Flexible materials should be used for the finger mounts and the mounting on the back of the hand to ensure comfortable long-duration sessions.

6. Size Adjustability

- a. Specification: The solution shall be able to fit hand lengths ranging from 5-9 inches.
- b. Rationale: The average male hand length is 7.6 inches and the average female hand size is 6.8 inches⁶. To accommodate this variability, the solution must be able to change length and width to fit most individuals.

7. Battery Life Span

- a. Specification: The solution shall last at least 3 hours of usage time.
- b. Rationale: A standard Meta Quest 3 provides 2-3 hours of battery life on a full charge⁷. To be comparable, the solution shall last the same usage time as the headset.

8. Hot Swappable Battery

- a. Specification: The battery must be replaceable without powering off the solution.
- b. Rationale: A hot-swappable battery provides maximum usage by preventing the need to wait for the product to charge. In addition, only the highest-end VR solutions have this ability, so bringing this to a cheaper size creates a serious market advantage.⁸

9. Hidden Cables

- a. Specifications: All wires should be fully encased and routed throughout the solution yet accessible for proper maintenance.
- b. Rationale (Self-Imposed): Exposed wiring causes the ability to snag and damage the product. To ensure the most seamless usage and the longest product life span, the cables shall be hidden yet accessible for maintenance.

10. Connectivity Types

- a. Specifications: The solution must have wired and wireless compatibility with the headset.
- b. Rationale: Wired connections can simplify troubleshooting issues, but a wireless connection is necessary for typical usage. In the market, Bluetooth and Wi-Fi are standard for connecting VR solutions to VR headsets. Products like bHaptics's

⁶ Sullivan, D. (2020, July 3). Average hand size: Adults, children, and relation to height. MedicalNewsToday. Retrieved December 12, 2024, from <https://www.medicalnewstoday.com/articles/average-hand-size>

⁷ Carter, R. (2024, May 6). How to Maximize Meta Quest 3 Battery Life: Top Tips. XRToday. <https://www.xrtoday.com/mixed-reality/how-to-maximize-meta-quest-3-battery-life-top-tips/>

⁸ "Weart TouchDIVER Pro Haptic Gloves." *Knoxlabs*, www.knoxlabs.com/products/weart-touchdiver-pro-haptic-gloves?_pos=1&_sid=51b4e4463&_ss=r. Accessed 12 Dec. 2024.

TactGlove utilize Bluetooth,⁹ while Weart's TouchDiver pro haptic glove utilizes both Wifi and Bluetooth.¹⁰

11. Manufacturing/Repairs and Part Supply

- a. Specifications: The solution must be made of modular replaceable components.
- b. Rationale (Self-Imposed): To keep manufacturing costs down and reparability viable, the product should be composed of 3D-printed parts and replaceable components. If something breaks the cost of repairing the product is minimized. For example, Velcro is a cheap and durable alternative to traditional zip ties or bolts, with a cycle life of around 20,000 uses.¹¹

⁹ bHaptic. (2024). *bHaptics TactGlove DK1 Wireless Haptic Gloves (Pair) - VR Accessory*. KnoxLabs. Retrieved 12 06, 24, from <https://www.knoxlabs.com/products/bhaptics-tactglove-dk1-haptic-gloves?srsId=AfmBOooZhpYmoK6P69NOkd7TPpjTSU4FyB2ZjaAdarD8ZvzBr73il8YI>

¹⁰ "Weart TouchDIVER Pro Haptic Gloves." *Knoxlabs*, www.knoxlabs.com/products/weart-touchdiver-pro-haptic-gloves?_pos=1&_sid=51b4e4463&_ss=r. Accessed 12 Dec. 2024.

¹¹ Ira, Stephen. "How Long Will It Stick? Understanding the Lifespan of Hook and Loop." *HookandLoop.Com*, 22 Nov. 2023, www.hookandloop.com/blog/how-long-will-it-stick-understanding-the-lifespan-of-hook-and-loop?srsId=AfmBOorzQca6ERDA6PkPlyNbKspVVyYw73iHR_iEGso-OZMVSNLgUUq_. Accessed 14 Dec. 2024.

II. Conclusion

To appeal to as many customers and stakeholders as possible, the proposed VR surgical solution shall satisfy the specifications listed above. The product heavily emphasizes a variable restrictive force and precise positional finger accuracy to create the most immersive experience possible, coupled with lightweight, moisture-resistant, and adjustable properties. Furthermore, a cost-effective product with replaceable parts allows for a more efficient distribution of high-quality medical training. Many of these specifications were determined through extensive research based on previous products' strengths and addressable weaknesses.